

Question Number	Answer	Mark
3(a)(i)	The only correct answer is C two <i>A is not correct because there is one maternal and one paternal chromosome with three gene loci</i> <i>B is not correct because there is one maternal and one paternal chromosome with three gene loci</i> <i>D is not correct because there is one maternal and one paternal chromosome with three gene loci</i>	(1)

Question Number	Answer	Additional guidance	Mark
3(a)(ii)	A description that makes reference to the following points: • uncoiling (of chromosome) (1) • used in protein synthesis / DNA replication (in S phase) (1)	Accept correct reference to histones / to form chromatin Accept transcription	(2)

Question Number	Answer	Additional guidance	Mark
3(b)(i)	• prophase I		(1)

Question Number	Answer Additional guidance	Mark
3(b)(ii)	The only correct answer is C three <i>A is not correct because ABe could not be formed by one chiasmata</i> <i>B is not correct because ABe could not be formed by one chiasmata</i> <i>D is not correct because ABe could not be formed by one chiasmata</i>	(1)

Question Number	Answer	Additional guidance	Mark
3(b)(iii)	An explanation that makes reference to two of the following points: - alleles b and e are closer together (1) - therefore {crossing over / chiasma} is less likely to occur between these alleles (1)	Accept converse for A and E Accept converse for A and E	(2)